

Chapter 11: Natural Resources of Nepal

Natural Resources of Nepal

The resources which exist naturally in the environment and can be exploited by humans to satisfy their needs or wants are called natural resources.

Natural resources can be classified based on their renewability as **renewable natural resources** and **non-renewable natural resources**.

In terms of the source of origin, they can be divided into **biotic** and **abiotic** natural resources.

Natural resources can also be categorized based on their stage of development as **potential**, **actual**, **reserve**, and **stock resources**.

Water Resources

Nepal is the second richest country in the world and the first in Asia in terms of water resource availability.

The main sources of water are **surface water**, **groundwater**, and **collected rainwater**.

Importance of Water Resources

The importance of water resources can be justified based on the following points:

1. Development of agriculture sector
2. Supply of energy
3. Development of the industrial sector
4. Development of the tourism sector
5. Development of transportation
6. Development of ICT
7. Healthy life of the citizen
8. Conservation of forest resources

Potentiality of Water Resources

The potentiality of water resources can be viewed in terms of the potentiality of **drinking water**, **irrigation**, and **hydroelectricity generation**, and other uses.

Current Status of Water Resources

The current status of water resources can be viewed in terms of **availability and use** of water resources. Based on the flow, water storage, and utilization, the rivers of Nepal can be categorized into three groups as the **first-level**, **second-level**, and **third-level** rivers.

The water resources of Nepal are being used in the forms of **drinking water**, **irrigation**, **hydroelectricity generation**, and **others**.

Problems of Water Resources

The problems of water resources in Nepal are associated with problems of **drinking water**, **irrigation**, and **hydroelectricity generation**.

Forest Resources

An area fully or partly covered by trees and shrubs is called a forest.

Current Situation of Forest Resources

The forest land covers **40.36 percent** and shrubs land covers **4.38 percent** of the total area of Nepal. Forest and shrubs land together represent **44.74 percent** of the total area of the country.

There are **35 major forest types** and **118 ecosystems** found in Nepal.

The 35 types of forests are classified into **10 broad groups** based on altitude.

The Government of Nepal has the policy to cover **40 percent** of the land area of Nepal by forest to maintain ecological balance.

Importance of Forest Resources

The importance of forest resources can be justified based on the following points:

1. Development of forest-based industries
2. Increase in employment opportunities
3. Support for agricultural production
4. Supply of energy
5. Promote forest tourism
6. Support livestock farming
7. Increase in foreign exchange earnings
8. Habitat for wildlife
9. Mitigate climate change and maintain ecological balance

Causes of Deforestation

The deforestation in Nepal is increasing day by day due to the following causes:

1. Overpopulation
2. Expansion of agriculture and livestock farming
3. Industrialization
4. Development activities
5. Over dependency on the forest for fuelwood

6. Mining
7. Illegal logging
8. Forest fires
9. Climate change
10. Lack of people's awareness

Consequences of Deforestation

The following are the major consequences of deforestation in Nepal:

1. Scarcity of fuelwood
 2. Increase in greenhouse gas emissions
 3. Disturbance in water cycle / desertification
 4. Soil erosion, landslide, and flooding
 5. Decrease in agricultural and livestock products
 6. Constraint to the development of forest-based industry
 7. Loss of species
 8. Reduction in exports and foreign exchange earnings
 9. Decrease in quality of life
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Mineral Resources

Mineral resources are the mineral deposits naturally available for exploitation and to satisfy the current or future needs.

Importance of Mineral Resources

The importance of mineral resources for the overall development of Nepal can be justified on the basis of the following points:

1. Fulfilment of necessities
2. Infrastructure development
3. Industrial development
4. Modernization of agriculture
5. Create employment opportunities
6. Increase in government revenue

7. Increase in foreign exchange earnings
8. Defence and warfare

Problems of Mineral Resources

The major problems of mineral resource development in Nepal are as below:

1. Lack of effective investor-friendly national mineral policy, mines and minerals acts and rules of the government
 2. Lack of investment
 3. Inadequate infrastructures
 4. Lack of energy
 5. Lack of highly skilled human resource
 6. Lack of scientific equipment and laboratory
 7. Illegal transactions of minerals
 8. Lack of mineral-based education and research
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Natural Resource Management, Climate Change and Its Effects on the Economy

Natural Resource Management

Natural resource management refers to the management of natural resources with a particular focus on how management affects the quality of life for both present and future generations.

The purpose of natural resource management is to improve the management of **land, forest, mineral, and water resources** to maintain both the productive capacity and ecological functions of these resources.

There is a direct relationship between **natural resource depletion** and **environmental pollution**.

The way of connecting links between environmental protection and the utilization of natural resources is **natural resource management**.

Climate Change

Climate change is the average condition for about thirty years. It refers to the kind of weather that's typically expected in a region over a longer period.

Effects of Climate Change on the Economy

The effects of climate change on the economy are as follows:

1. Environmental degradation
2. Impacts on agriculture

3. Impact on economic growth
4. Impacts on health
5. Impact on energy and water demand
6. Impact on tourism
7. Impact on the ecosystem